

Woodward (2026)– “Explanation, Truth, and Structural Realism” Pt. 1

What is the role of truthlikeness in providing successful explanations?

In this paper, Woodward sets out to articulate a framework, in which select accounts of explanation and scientific (structural) realism link up with truthlikeness in a mutually supportive way.

- Philosophers have disagreed about what kind of truth requirements are central to the effectiveness of scientific explanations.
 - Proponents of entity realism have suggested the theory must pick out the right ontology.
 - David Wallace has championed a form of math-first structural realism, in which the latching on to the correct mathematical relations in the world enables successful explanation and representation in physics.
- One sense in which a theory may be truthlike is if it is **an effective theory**– a good approximation to some more general theory (which need not yet be known).
- Woodward’s idea of approximation/ effectiveness/ truthlikeness “is [inherently] subject-matter specific in the sense that it depends on the particular theories under discussion and particular claims those theories make” (p.1, footnote).
- Truthlikeness cannot be given a general characterisation that is independent of any discipline.

The W-account: What if things had been different? What enables successful explanation is the truthlikeness of the explanation’s structural claims.

1. **Q:** What is a version of realism that fits well with the w-account? (p. 1) Given a plausible version of realism, what account of explanation fits best with it? (p. 1)

A: The w-account in conjunction with a version of structural realism fit best and provide the best accurate description of scientific practice.

2. **Q:** What in an explanation needs to be truthlike for it to successfully explain?

A: The dependency relations need to be at least approximately right within the explanation's domain of application. So do the quantities that serve as inputs to those relations.

Dependency relation: lawful or causal relations characterising how the values of variables depend on others.

Examples:

- In Newtonian gravitation, the gravitational force between two bodies *depends* on their masses and the distance between them.
- The effectiveness of SSRIs (selective serotonin reuptake Inhibitors) *depends* on their ability to block the reabsorption of serotonin in the brain.
- The rate of a chemical reaction *depends* on the temperature and surface area of the reagents.

What kind of Truthlikeness?

Not the IBE kind.

The w-account enacts a truthlikeness condition on explanation.

1. Unlike IBE, the w-account does not rank potential explanations wrt their goodness if true.
2. Nor does it connect "potential explanatoriness" with truth, nor reason to believe.

Woodward criticises IBE for supposing that we can infer truth from potential explanatoriness. He argues that successful explanation turns on the appropriate sort of truthlikeness in the explanans.

Establishing that something is an explanation must involve independent evidence that it is truth-latching, *independent from* its explanatory potential.

Grievance: Realism is frequently lauded as the best **explanation** for the success of science. But by *which* definition of **explanation** this claim may be true is a neglected question.

“Views about explanation and claims about what version of realism is most defensible should mutually constrain each other” (p. 3).

If we wish to say that science explains because it picks out the correct (structural) dependency relations, science should provide true information about said dependency relations.

An Eliminative Induction Schema

What might it look like for there to be *independent evidence supporting the truthlikeness of (the appropriate parts of) an explanation?*

Schema S

1. Explanandum E has some explanation.
2. The explanation of E , whatever it may be, must satisfy a truthlikeness requirement
3. $Ex_1, \dots, Ex_n$ are mutually exclusive potential explanations of E . Moreover, this list is exhaustive. (Ex_i are consistent with current background knowledge and evidence.)
4. We generate additional evidence that excludes (falsifies, undermines, renders implausible) all of the potential explanations in (3), except for one: Ex_k .
5. Ex_k is truthlike. (1-4)
6. Ex_k explains E . (1-5)

Schema S embodies Woodward’s requirement that Ex_i be independently truthlike, separate from its potential to explain.

Schema S mandates inference to the only remaining explanation. In doing so, it makes a handful of suppositions.

- It assumes E has some explanation (in the sense understood by the w-account) (premise 1).

- It excludes “as if” explanations (premise 2).
- Without the established connection between truthlikeness and explanation in premise 2, we would not be able to infer 5.
- **Schema S** does make assumptions about explanation as it related to truth, but no steps take explanatory potential to be a proxy for truth.

There are a handful of ways step 4 could go and Woodward wants to be permissive:

1. We may be able to deduce the falsity of all but one Ex_n given accepted background assumptions and evidence.
2. The evidence could render all but one Ex_n implausible or improbable via an eliminativist Bayesian strategy.
3. We may be able to rule out many Ex_n as implausible coincidences.
4. Some Ex_n may not be inconsistent with established facts, but fail to explain them nonetheless, whilst Ex_k does.

The W-Account of Explanation and Its Implications for Explanatory Assessment

“... the key idea of the w-account is that explanations work by providing what-if-things-had-been-different information about explananda– by citing factors in their explanans such that if those had been different in various ways, the explananda would have been different and by citing the patterns of dependence of the explananda on the factors cited in the explanans.” (p. 7)

- Dependency relations are the answer to w-questions, which encode claims about counterfactuals interpreted in an interventionist style.
- Woodward’s account is committed to a type of minimal modal realism.
- In successful explanations, the explananda need to depend on factors of the explanans in a truthlike manner.

- Newtonian gravitational theory provides effective explanations for a variety of explananda.
- According to the w-account, its explanatory power is not undermined by its limited domain of application.
- Rather, that the dependency relations postulated by the theory are truthlike within its domain “establishes its explanatory bona-fides” (p. 7).

Why bother?

- We may wonder: why both with understanding explanation– do we even have explanations in science?
- We’d like to think so, but even if we *didn’t*:
 - The w-account allows our assessment of what we *do know* to influence our view of what we should expect from successful explanations.
 - A lot of what we know seems structural.
 - The vast majority of our theories in science are effective theories.
 - The w-account, being highly interventionist, can work with this because it does not require a full-fledged theory of everything to offer an account of what makes explanation go.

Comparing Potential Explanations with the W-account

Under the w-account, we can compare some types of potential explanations, but not all. We cannot compare the kind on which IBE rely.

What can the w-account compare?

- Cases when Theory 2 correctly answers more w-questions and explains more explananda than Theory 1. (e.g. Newtonian gravitational theory and General Relativity)
- Theory 1 provides correct answers to qualitative w-questions and Theory 2 provides the same answers, but also more quantitative detail.

What can't the w-account compare?

- Unlike IBE, the w-account does not license comparisons between explanations on the basis of virtues such as *simplicity* or *unification* (p. 9).
- Rather, if a class of explanations account for the same explananda equally well despite varying levels of complexity, further information is required to discriminate amongst them.
- Non-explanatory virtues do not feature in the w-account's picture of the relationship between explanation and truth.
- Moreover, Woodward argues that IBE-style inferences are deeply contrary to the established norms of scientific practice.
 - Example: explanations that identify confounding variables would be seen as, *ceteris paribus*, less credible.
 - In practice, to reduce the credibility of comparatively complex explanations, we need independent evidence to dis-confirm them.
 - IBE-style standards of explanation reward “virtues” that are not reliably correlated with explanatory power.

References

Woodward, James (2026). Explanation, truth and realism. *Synthese* 207 (1):14.